DIGITAL BRIDGE FOR EMERGENCY RESPONSE IN RURAL MADURAI

[1]Sahithya P
Department of Computer Science of Engineering
Thiagarajar College of Engineering, Madurai, Tamilnadu, India
sahithya@student.tce.edu

[1]Chandana V
Department of Computer Science of Engineering
Thiagarajar College of Engineering, Madurai, Tamilnadu, India
chandana@student.tce.edu

[1]Farjana G
Department of Computer Science of Engineering
Thiagarajar College of Engineering, Madurai, Tamilnadu, India
farjana@student.tce.edu

[2]Sunija A J
Department of Chemistry
Thiagarajar College of Engineering, Madurai, Tamilnadu, India
ajschem@tce.edu

[3]Dharani J
Department of Computer Science of Engineering
Thiagarajar College of Engineering, Madurai, Tamilnadu, India
jdicse@tce.edu

[3]Pradeepkannan D
Department of Electronics and Instrumentation Engineering
K.L.N College of Engineering, Pottapalayam, Sivagangai Dist, Tamilnadu, India
pradeepkannand@gmail.com

Abstract - First-aid interventions for medical emergencies during the critical golden hour continues to be a major challenge especially in rural areas where access to knowledge of first-aid, access to professional medical services near-by, and referral to tertiary hospitals are limited. In Usilampatti, Madurai District of India, there is a shortage of trauma specialists and low levels of first aid awareness leading to delays in treatment and increased likelihood of developing non-preventable complications. To fill this gap, the mobile app, Aarudhra was developed to offer rural users with a user-friendly mobile application providing access to offline first-aid guidance when required. The application offers access to guidance in Tamil, an easy-to-use interface, and simple graphic and image-based buttons. The application was designed for rural users with limited literacy and can provide guidance and resources without requiring continuous internet access. The mobile app contains an emergency call option. It can locate hospitals for the user, has a chatbot function for users to find answers to questions about common first aid issues, contains tools to monitor user health, medication reminders, and provides written and spoken instructions regarding common first-aid emergencies such as cuts, burns, strokes, snakebites, fever, and headache. The mobile app was developed using Kodular and MIT App Inventor, using a structured methodology including algorithms, system design, module development and testing. The application was created with validated guidelines from various resources such as the WHO and Red Cross, so users have an increased chance of acting successfully when emergencies occur. It also provides awareness of health issues in rural areas and supports Sustainable Development Goal 3 (Good Health and Well-Being) by helping to lessen the impact of delays in receiving emergency medical care and therefore increases first-aid knowledge in rural communities.

Keywords - First-aid awareness, Emergency healthcare systems, Android application, Rural medical assistance, Human-centered design, Sustainable Development Goals

I. INTRODUCTION

Timely first-aid favour during a medical emergency can save a life, it is most critical in rural areas where people don't have easy access to specialist healthcare. In places like Usilampatti, the patients suffer more because of the delay in the emergency response and the lack of awareness. Most people in the community don't know how to do the basic lifesaving things and the first-aid stuff that they do have is mostly in the form of text, in English, or through the internet, so it's not really suitable for low-literate, Tamil-speaking people. This research work, thus, comes up with a mobile application, Aarudhra, that gives first-aid guidance in the Tamil language through simple text, audio instructions, and visual aids without the need for internet. Besides the first-aid part, the application is equipped with facilities like emergency contacts, hospital locators, health monitoring system, simple chatbot and medication reminders. Aarudhra helps the rural people to be better prepared and equipped at the first and most crucial moment of an emergency by overcoming language, literacy, and accessibility barriers. This kind of community-level intervention brings down the incidence of those complications which have been a result of negligence and at the same time is a step towards the realization of the good health and well-being of people living in the deprived areas.

II. LITERATURE REVIEW

Rural Emergency Healthcare and the Imperative for Community-Based First-Aid Interventions:
According to Saravanan M.P. [1], who wrote for The New Indian Express, there has been a spike in the number of trauma patients being transferred from Usilampatti Government Hospital to tertiary care centers because of the absence of specialist doctors and the resulting delays during the 'golden hour'. This points towards a real need for immediate access to first-aid care at the community level. The Directorate of Medical and Rural Health Services, Tamil Nadu [2] found that, while essential life-saving material is available in government hospitals, there are virtually no properly trained trauma specialists available to provide timely intervention and hence further supports the need for pre-hospital first-aid intervention. Thennathayalan S and Pugazhentiran M. [3] more specifically conducted a comparative study of the availability of first-aid supplies at different sites within MGNREGS; they found insufficient numbers of basic supplies, as well as many of the rural workers were not prepared to deal with medical emergencies. The Census of India - Usilampatti Data [4] indicates that this region of Tamil Nadu has a population of over 2.09 lakh with significant variability in their educational level; this further reinforces the need for accessible, easy-to-understand and usable first-aid in regional language like Tamil. According to Singh K. et al. [5] they assess how emergency medical services operate in rural India but identify delays in ambulance response as well as shortages of trained emergency medical professionals negatively impacting the outcome of patients. The World Health Organization [6] has established evidence-based first-aid practices with standardized emergency care guidelines including snake bite treatment while mentioning specific traditional practices to be avoided and emphasizing the use of modern emergency care systems. The International Federation of Red Cross and Red Crescent Societies [7] provides internationally accepted first-aid procedures and the Indian Red Cross Society [8] modifies those procedures taking Indian rural conditions into account, so they can be effectively utilized by the rural population. The Meghalaya State Disaster Management Authority [9] has developed standard first-aid instructions that incorporate visuals and language appropriate for low-literate readers. The Indian Council of Medical Research [10] provides clinical guidance regarding danger signs of emergency medical conditions requiring immediate medical evaluation. The Indian Medical Association [11]

provides the public with information about common presentations like headaches and their associated danger signs. The Ministry of Health and Family Welfare [12] and National Rural Health Mission [13] have both conducted nationwide assessments of the health system including identifying shortages of healthcare professionals and emergency services across rural regions. According to The World Health Organization [14], when pre-hospital trauma care efforts are organized (such as access to early first aid), use of rapid transport, and presence of trained individuals as first responders, there is a significant decrease in the number of preventable deaths / disabilities occurring during emergency situations (particularly in low-resource communities or rural areas). Imron Subhan and Anunaya Jain., [15] discusses how developing basic infrastructure for any type of emergency service is a necessity for improving survival rates in rural and impoverished regions throughout India. Examples of this include building strong community help or first aid systems and establishing a mechanism through which people can receive pre-hospital care immediately after an incident occurs. The National Sample Survey Office [16] has reported that healthcare access is significantly lower for both rural and urban areas. Bledsoe, B.E., [17] has shown that first-aid provided by laypeople in a timely fashion will reduce complication rates and mortality rates. The Ministry of Labour & Employment [18] believes that worksite first-aid preparedness is an essential element of emergency medical services; however, there are still substantial gaps between policy and practice. UNICEF India, Communication for Development [19] states that utilizing local languages, images, and simplified information when communicating about health can greatly increase understanding amongst low educational attainment individuals. Clinical protocols provided by the All India Institute of Medical Sciences [20] emphasize the importance of early intervention through pre-hospital care and early stabilization for improving emergency medical care outcomes. Rahman A., et al., [21] mentioned the use of mobile health programs significantly creates improved understanding and response to emergencies. According to National Disaster Management Authority (NDMA), Govt. of India [22], it is imperative to promote early response as well as building community resilience when faced with disasters/injuries. Ghosh S., et al., [23] conducted a study demonstrating that an increased severity of workplace-related injuries is associated with an increased risk of injury because of a lack of workplace knowledge and preparedness. Therefore, basic first-aid training should be available to the general public. The World Bank's report ‘Rural Health and Health System Strengthening in India’ [24] describes the lack of access to emergency health care in rural areas and presents evidence that low-cost, digital and community-based health care interventions can improve early response and decision making. Research conducted in the Journal of Telemedicine and Telecare by Smith et al., [25] shows that the use of mobile health and telemedicine technology increases awareness of emergency situations, improves the speed of response to emergencies, and improves clinical decision making in rural populations.

Existing First-aid and Healthcare Apps and Studies:

The First Aid IFRC app [26] provides standard emergency guidance prepared by the International Federation of Red Cross, however, it has very limited localization features and only minimal offline interactivity. The ESTEPS First Aid app [27] has a nice and easy user interface presenting basic first-aid instructions, yet it is missing audio support and advanced emergency workflows. iFirstAid [28] has information about survival and responding to emergencies but it is mostly text-based and not very convenient for rural users. Baby and Child First Aid [29] by the British Red Cross is mainly about children's emergencies and has limitations as to the age group. UN Buddy First Aid

[30] has general emergency awareness content but its big size and low offline usability make it less accessible in low-resource environments. From the study of existing first-aid apps, it can be seen that none of them offer extensive support in regional languages like Tamil. The lack of text and audio guidance in the local language is a major limiting factor in how effective these solutions can be for the rural and non-English speaking populations.

III. METHODOLOGY

System Architecture and Design Methodology:
The Aarudhra mobile application [31] is a first-aid assistance that is quick, dependable, and easy-to-use which is used especially during situations of emergencies when the people cannot get immediate help from medical professionals, and there is no stable internet connection accessible. It is built with an offline-first architectural approach in mind so that the running, first-aid instructions in text and picture formats, Tamil audio support, and emergency-related information are all deeply integrated within the app thus ensuring that the app remains usable without any interruption, no matter if there is no network available. The app architecture is divided into two main layers, i.e. the interface layer, and the application logic layer. Through big visual images, clearly written buttons, and simple Tamil words with text-to-speech support, the interface layer has been made to help users have a hassle-free experience and it thus caters to users of different levels of literacy. The application logic layer handles user actions like choosing the emergency type, moving through different modules, turning on audio instructions, changing health parameters, and setting reminder alerts, thereby maintaining the flow of the work and smoothness of the operation.

Operational Workflow and Functional Modules:
The Aarudhra app starts with a splash screen setting the tone of the app which is then followed by the home dashboard that acts as the main navigation centre for all the major features. The home dashboard lets the users reach out to the core modules such as First Aid, Hospital Locator, Chatbot, and Health Monitoring that are all designed for clear and intuitive navigation even in the case of deep-level features. Through the First-Aid module, one can go for different emergency cases such as a cut, burn, fever, snakebite, headache, or stroke, and the app instantly fetches the relevant offline material and along with the visual demonstration, it also conveys step by step instructions through Tamil audio which can be accessed through text-to-speech technology. Such a solution makes it possible to get accurate and quick help even where resources are scarce. The Hospital Locator tool makes use of GPS through the OpenStreetMap service to locate and pinpoint the position of a hospital near the user, thereby making it possible for the user to get professional medical care after the provision of first aid at home. The Chatbot package has pre-recorded answers to the common health questions that are medically reviewed and it also supports the Tamil audio format thus, it gives strictly controlled, educative, and interactive help without being connected to the internet continuously. The Health Monitoring module allows users to input and keep a record of their health parameters such as body mass index, blood pressure, blood sugar, heart rate, and weight, also it gives medicine reminder functionality through scheduled notifications. All health data input by the user are saved only on the device and displayed in the form of simple graphical charts for easy understanding and long-term tracking.

Implementation Strategy, Tools, and Testing Process:
The application was created with block-based platforms like Kodular and MIT App Inventor, which were chosen because they are lightweight, allow for very rapid prototyping, and are compatible with low-end Android devices that are generally used in rural areas. Different supporting parts and services, such as text-to-speech engines,

notifier modules, chart components, web viewers, and OpenStreetMap APIs, were combined to improve functionality while still maintaining efficiency and low resource consumption. The implementation plan focused on a clean and minimal user interface, full Tamil language support, offline capability, and consistent navigation by including a home icon on every screen. Careful functional testing, usability testing, offline testing, performance on low-end devices, and systematic bug fixing correspond to a well-structured test methodology. By following this methodical and well-thought-out implementation process, the Aarudhra app is exemplary in meeting its goal of being a reliable, accessible, and culturally appropriate source of first-aid and health support for the rural population.

IV. RESULTS AND DISCUSSIONS

The Aarudhra app is designed to empower people living in rural or isolated areas who may not have easy access to medical help, to get a full range of first-aid services. The app serves as a guide for users on how to manage various medical emergencies like cuts, burns, snake bites, stroke symptoms, fever, and headaches. It offers features such as offline first aid guidance, one-touch emergency calling, hospital locator, health chatbot, medication reminders, health monitoring system, and also text-to-speech instruction delivery. Moreover, the app's user interface is quite straightforward and geared towards accessibility, in that it employs icons, pictures, and sound cues to facilitate users with limited literacy or low medical knowledge to comprehend how to use the app.

User Interface Modules and Application Screens:
(i) The home screen of the Aarudhra app is the primary means of navigating through the app and has all four key modules readily accessible to users from the main page. The 108-emergency call button is readily visible, allowing for immediate access to an ambulance in emergency situations. The icon-based layout allows for ease of access to first-aid instructions, hospital locator services, chatbot support, and health-monitoring functions with graphical tracking.

Fig.1 Splash Screen and Home Screen

(ii) The first-aid guidance screen lists general instructions for treating common injuries and illnesses (e.g. cuts, burns, snake bite, fever, headache, stroke). For each of these injuries and illnesses, the instructions are given using words, pictures, and spoken-word audio so the user can follow the steps for each instruction easily. Providing all of these for each injury or illness enables the user to understand very quickly and easily even if they are illiterate or do not have much medical knowledge.

Fig.2 First-Aid Screen

(iii) The hospital locator screen enables the user to find the nearest hospital or healthcare facility. This screen also provides navigation assistance to the user. In addition to the navigation assistance, there

are one-touch emergency call options to both 101 (fire and rescue) and 108 (ambulance) to facilitate quicker coordination amongst emergency responders and quicker access to professional assistance.

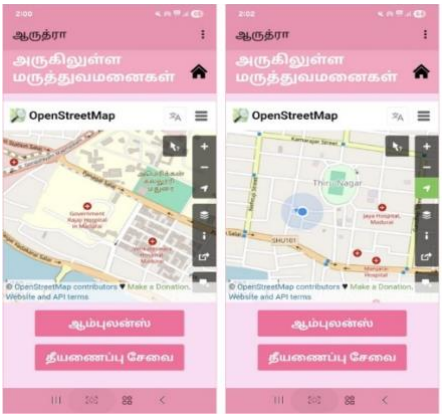

Fig.3 Hospital Locator Screen

(iv) The chatbot assistance screen is available to the user to provide interactive assistance for responding to questions related to emergencies and questions about basic health concerns.

Fig.4 Chatbot Screen

(v) The health monitoring screen is a type of screen that provides people with assistance in terms of preventative healthcare and managing their health on an ongoing basis, which includes:

1) Calculating BMI (Body Mass Index);
2) Creating medicine reminders;
3) Keeping track of blood sugar, blood pressure, and Heartrate using line charts.

Fig.5 Health Monitoring Screen

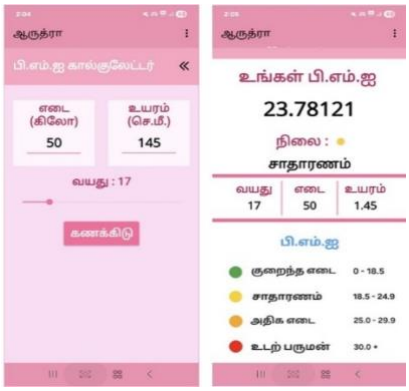

Fig.6 BMI Calculator Screen

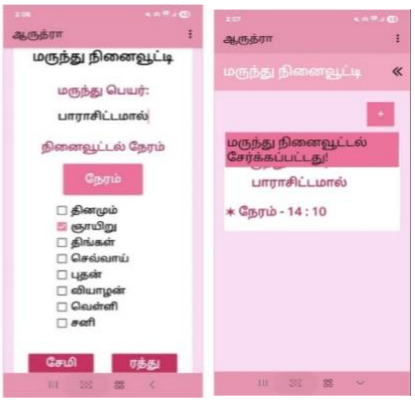

Fig.7 Medicine Reminder Screen

Fig.8 Health Tracker Screen

Functional Testing:
All core functional areas underwent thorough functional testing. Functional testing was carried out for validation of access to first aid instructions, emergency calls and access to hospital location (GPS), responses from the chatbot, reminding users to take medications and tracking their health, and the ability of the app to speak to users. All functional areas were tested during development and operational (live) phases to verify their correct operation and stability of the system. Any bugs observed in the software were recorded, evaluated and addressed to allow for smooth interaction and no errors.

Accessibility Assessment:
The purpose of the accessibility assessment was to ensure that people with low literacy levels/limited experience using a smartphone could successfully use the app. The use of icons, images and audio instructions decreased reliance on written text so that all functions of the app can be easily used in an intuitive manner, increasing inclusion. Additionally, the text to speech feature increased accessibility by giving users the option of hearing instructions as opposed to reading them.

V. CONCLUSION

The Aarudhra application exemplifies utilizing technology to save lives by providing rural health system access. Rural communities may have little or no medical facilities available, so many do not know how to react when faced with an accident or an unexpected health crisis/condition during an emergency. The Aarudhra application provides easy-to-access, quick, and simple first-aid instructions even if an Internet connection is not available, making it a viable resource in rural settings. Additionally, the application uses simple symbols and pictures to provide both Tamil language medical instructions for individuals who may be unable to read or write or those who speak Tamil but have no English language knowledge. Aarudhra is a multi-functional app by also allowing users to save their emergency contacts, monitor their overall health, find a hospital, and provide medication reminders. By offering these multiple functions, Aarudhra is shifting from an 'emergency usage' application to a 'daily health companion' application. Overall, Aarudhra is a tool that promotes effective preparation, confidence, and community health. It will lay the foundation for a regional language-specific health-care delivery system that can be utilized throughout India.

VI. REFERENCES

[1] M. P. Saravanan, “Shortage of specialist doctors delays trauma care in rural Tamil Nadu,” *The New Indian Express*, Chennai, India, 2021.

[2] Directorate of Medical and Rural Health Services, Tamil Nadu, *Status Report on Rural Emergency Healthcare Facilities*, Government of Tamil Nadu, Chennai, India, 2020.

[3] S. Thennathayalan and M. Pugazhentiran, “Assessment of first-aid preparedness among MGNREGS workers in rural areas,” *Indian Journal of Occupational Health*, vol. 62, no. 3, pp. 215–221, 2019.

[4] Census of India, *District Census Handbook: Usilampatti*, Government of India, New Delhi, India, 2011.

[5] K. Singh, R. Sharma, and P. Verma, “Emergency medical services in rural India: Challenges and delays,” *Journal of Emergency Medical Services in Developing Regions*, vol. 5, no. 2, pp. 45–52, 2018.

[6] World Health Organization, *Guidelines on Basic Emergency Care: Approach to the Acutely Ill and Injured*, WHO Press, Geneva, Switzerland, 2018.

[7] International Federation of Red Cross and Red Crescent Societies, *International First Aid and Resuscitation Guidelines*, IFRC, Geneva, Switzerland, 2020.

[8] Indian Red Cross Society, *First Aid Manual for Indian Conditions*, Indian Red Cross Society, New Delhi, India, 2019.
[9] Meghalaya State Disaster Management Authority, *Community-Based First Aid Training Manual*, Government of Meghalaya, Shillong, India, 2017.
[10] Indian Council of Medical Research, *Clinical Guidelines for Emergency Medical Conditions*, ICMR, New Delhi, India, 2020.
[11] Indian Medical Association, *Public Awareness Guidelines on Common Medical Emergencies*, IMA Publications, New Delhi, India, 2019.
[12] Ministry of Health and Family Welfare, *National Health Profile*, Government of India, New Delhi, India, 2021.
[13] National Rural Health Mission, *Rural Health Statistics in India*, Government of India, New Delhi, India, 2020.
[14] World Health Organization, *Prehospital Trauma Care Systems*, WHO Press, Geneva, Switzerland, 2017.
[15] I. Subhan and A. Jain, "Improving survival rates through community-based emergency infrastructure in rural India," *International Journal of Public Health Systems*, vol. 6, no. 1, pp. 30–37, 2019.
[16] National Sample Survey Office, *Health in India: NSS 75th Round*, Ministry of Statistics and Programme Implementation, Government of India, New Delhi, India, 2018.
[17] B. E. Bledsoe, "The role of bystander first aid in reducing mortality," *Prehospital Emergency Care*, vol. 14, no. 4, pp. 437–445, 2010.
[18] Ministry of Labour and Employment, *Occupational Safety and First Aid Preparedness Guidelines*, Government of India, New Delhi, India, 2019.
[19] UNICEF India, *Communication for Development: Health Messaging in Low-Literacy Communities*, UNICEF India Country Office, New Delhi, India, 2018.
[20] All India Institute of Medical Sciences, *Emergency Medicine Clinical Protocols*, AIIMS Publications, New Delhi, India, 2020.
[21] A. Rahman, S. Kumar, and M. Patel, "Impact of mobile health applications on emergency response awareness," *International Journal of Medical Informatics*, vol. 129, pp. 1–7, 2019.
[22] National Disaster Management Authority, *National Guidelines on Medical Preparedness and Emergency Response*, Government of India, New Delhi, India, 2016.
[23] S. Ghosh, R. Banerjee, and A. Das, "Workplace injury severity and first-aid preparedness in developing regions," *Journal of Occupational Safety*, vol. 11, no. 2, pp. 85–92, 2018.
[24] World Bank, *Rural Health and Health System Strengthening in India*, World Bank Publications, Washington, DC, USA, 2019.
[25] J. Smith, L. Brown, and R. Taylor, "Mobile health and telemedicine for emergency response in rural populations," *Journal of Telemedicine and Telecare*, vol. 26, no. 5, pp. 289–296, 2020.
[26] International Federation of Red Cross and Red Crescent Societies, *First Aid IFRC Mobile Application Documentation*, IFRC, Geneva, Switzerland, 2021.
[27] ESTEPS Consortium, *ESTEPS First Aid Mobile Application Overview*, European Training Program for Emergency Services, 2020.
[28] iFirstAid Development Team, *iFirstAid Application User Guide*, iFirstAid Publications, 2019.
[29] British Red Cross, *Baby and Child First Aid Application Manual*, British Red Cross Society, London, UK, 2020.
[30] United Nations Development Programme, *UN Buddy First Aid Application: Concept and Features*, UNDP Publications, New York, USA, 2021.
[31] Sahithya P, Chandana V, Farjana G., "AARUDHRA" GitHub repository, 2026.
Source Link: https://github.com/sahithya-5508/AARUDHRA/blob/main/AARUDHRA.apk